

Preventive Maintenance SOP

This SOP defines the routine maintenance schedule and the required operator response when a machine sensor reading crosses a WARNING or DANGER threshold as defined in the Machine Operation Manual.

Section 1: Routine Maintenance Schedule

Every machine in the fleet receives a scheduled inspection every 168 operating hours (approximately weekly under continuous 3-shift operation). The inspection covers lubrication, belt/bearing wear, sensor calibration, and a visual check for leaks or loose fasteners. Inspection results are logged in the maintenance record with machine ID, timestamp, and inspector ID.

Section 2: Response to a WARNING-level reading

If any sensor channel (temperature, vibration, pressure, or torque) reaches its WARNING threshold and remains there for more than one full shift, the operator must: (1) log the reading and machine ID in the maintenance system, (2) increase monitoring frequency for that machine to once per hour, and (3) notify the shift supervisor. The machine may continue running under increased monitoring; a WARNING reading alone does not require stopping the machine.

Section 3: Response to a DANGER-level Reading

If any sensor channel reaches its DANGER threshold, the operator must treat this as a priority maintenance event, not a routine log entry. Specifically for dangerously high vibration: (1) reduce the machine's load or production rate immediately to lower mechanical stress, (2) schedule the machine for preventive maintenance within the current shift rather than waiting for the next scheduled inspection, (3) do not wait for a full failure before intervening — a sustained DANGER-level vibration reading is a strong precursor to bearing or shaft failure, and (4) record the event, the action taken, and the responsible supervisor in the maintenance log. If the DANGER-level reading is accompanied by unusual noise, visible smoke, or a burning smell, follow the emergency stop procedure in the Safety & Emergency Guidelines document instead of continuing to operate the machine at reduced load.

Section 4: Escalation

Any DANGER-level event that recurs on the same machine within 72 hours after a maintenance action must be escalated to the maintenance engineering team for root-cause investigation, rather than repeating the same corrective action.